

Analysis of Salary Data

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Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Regression Results
- Interpretation and Next Steps

Data Description

- In the dataset we had **373** employees.
- Two fully empty observations discovered and omitted.
- Max. salary was \$ **250000**.
- Min. Salary was \$ **350**.
- Median salary was \$ **95000**.

Data Description



- **The salary distribution graph seems to be a right skewed one.**

Data Description



- A strong positive correlation between `Years of Experience` and `Salary`.

Regression Models

- **Sklearn** and **Statsmodels** (modules: sm & smf) libraries used.
- statsmodels **smf** & **sm** considered to solve the project's questions.
- R-Squared values were very good for a linear model:
 - Sklearn -> **0.900**
 - Statsmodel(s) -> **0.915**

Regression Results

OLS Regression Results

Dep. Variable:

Salary

R-squared:

0.915

Model:

OLS

Adj. R-squared:

0.913

Method:

Least Squares

F-statistic:

391.7

Date:

Wed, 10 Dec 2025

Prob (F-statistic):

2.41e-187

Time:

15:32:40

Log-Likelihood:

-4090.6

No. Observations:

373

AIC:

8203.

Df Residuals:

362

BIC:

8246.

Df Model:

10

Covariance Type:

nonrobust

coef

std err

t

P>|t|

[0.025

0.975]

Intercept

3.212e+04

2485.604

12.921

0.000

2.72e+04

3.7e+04

is_director[T.True]

2.539e+04

3535.194

7.181

0.000

1.84e+04

3.23e+04

is_junior[T.True]

-4492.5634

2458.516

-1.827

0.068

-9327.330

342.203

is_senior[T.True]

1.307e+04

1976.833

6.610

0.000

9178.417

1.7e+04

is_manager[T.True]

4434.5081

1960.352

2.262

0.024

579.400

8289.617

is_analyst[T.True]

-176.3409

2103.878

-0.084

0.933

-4313.698

3961.016

is_engineer[T.True]

2362.9515

3561.972

0.663

0.508

-4641.805

9367.708

Q('Education Level')[T.Master's]

1.345e+04

2053.896

6.547

0.000

9408.638

1.75e+04

Q('Education Level')[T.PhD]

2.327e+04

2801.927

8.304

0.000

1.78e+04

2.88e+04

gender_male

7484.7304

1509.904

4.957

0.000

4515.446

1.05e+04

Q('Years of Experience')

4937.4278

197.548

24.994

0.000

4548.941

5325.914

Omnibus:

130.784

Durbin-Watson:

1.815

Prob(Omnibus):

0.000

Jarque-Bera (JB):

1091.992

Skew:

1.230

Prob(JB):

7.53e-238

Kurtosis:

11.013

Cond. No.

63.1

- R-squared value is 0.915

Regression Results

- In our ``smf.ols`` summary, look at the ``P>|t|`` column for each coefficient.
- A common rule: a feature is significant if its p-value is less than 0.05 (or our chosen α).
- From our printed table, significant ($p < 0.05$) features are:
 - ``is_director[T.True]``
 - ``is_senior[T.True]``
 - ``is_manager[T.True]``
 - ``Q('Education Level')[T.Master's]``
 - ``Q('Education Level')[T.PhD]``
 - ``gender_male``
 - ``Q('Years of Experience')``
- Not significant ($p \geq 0.05$) in the first model:
 - ``is_junior[T.True]`` ($p \approx 0.068$)
 - ``is_analyst[T.True]`` ($p \approx 0.933$)
 - ``is_engineer[T.True]`` ($p \approx 0.508$)

Regression Results

OLS Regression Results						
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Dep. Variable:	Salary	R-squared:	0.914			
Model:	OLS	Adj. R-squared:	0.913			
Method:	Least Squares	F-statistic:	557.4			
Date:	Wed, 10 Dec 2025	Prob (F-statistic):	1.46e-190			
Time:	15:42:44	Log-Likelihood:	-4092.6			
No. Observations:	373	AIC:	8201.			
Df Residuals:	365	BIC:	8233.			
Df Model:	7					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	2.876e+04	1677.411	17.146	0.000	2.55e+04	3.21e+04
is_director[T.True]	2.498e+04	3503.028	7.131	0.000	1.81e+04	3.19e+04
is_senior[T.True]	1.445e+04	1820.427	7.936	0.000	1.09e+04	1.8e+04
is_manager[T.True]	4284.6595	1822.582	2.351	0.019	700.579	7868.740
Q('Education Level')[T.Master's]	1.404e+04	2032.078	6.908	0.000	1e+04	1.8e+04
Q('Education Level')[T.PhD]	2.308e+04	2760.740	8.360	0.000	1.77e+04	2.85e+04
gender_male	7629.6521	1481.834	5.149	0.000	4715.649	1.05e+04
Q('Years of Experience')	5104.4388	177.226	28.802	0.000	4755.927	5452.950
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Omnibus:	126.635	Durbin-Watson:	1.790			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1010.895			
Skew:	1.196	Prob(JB):	3.07e-220			
Kurtosis:	10.702	Cond. No.	60.8			
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Fit the reduced model

- Drop only the non-significant terms from the formula and refit:
- Now all remaining predictors should be significant in the new model.
- We can compare R^2 / Adj. R^2 between the full and reduced models and note that we didn't loss much explanatory power.

Interpretation and Next Steps

$$\begin{aligned}\text{Expected_Salary} = & 28760.8 + 7629.7 \cdot \text{gender_male} + 24980.8 \cdot \text{is_director} + 14446.5 \cdot \text{is_senior} \\ & + 4284.7 \cdot \text{is_manager} + 14037.3 \cdot \text{Education_Level_Master} \\ & + 23080.8 \cdot \text{Education_Level_PhD} + 5104.4 \cdot \text{Years_of_Experience}\end{aligned}$$

- On average, a person with a Senior title is expected to earn \$14,446 more than a person with no director, senior, or manager title, assuming they have the same gender, years of experience, and education level.
- The 95% confidence interval for the Education Level_PhD coefficient is 17,700, 28,500 USD (approx), meaning we are 95% confident that the true salary premium for a PhD holder lies within this range.